BME205 – Fundamentals of Biomedical Engineering

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With Exam Question Callouts

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1 Homeostasis & Negative Feedback

Whole-body homeostasis maintains stable internal conditions using a negative feedback loop. Key variables that the body regulates include: body temperature, blood glucose, blood pH, blood pressure, and ion concentrations (e.g., K^+, Na^+, Ca^{2+}). These are called **regulated variables**.

1.1 Main Control Pathway

1.1.1 Control Pathway Components

a. **Reference point (set point)** — the desired value of a regulated variable. Example: body temperature $\approx 37°C$.

b. **Sensor (Receptor)** — detects changes in the regulated variable. Example: thermoreceptors in the skin.

c. **Afferent neurons** — carry sensory information from the sensor to the brain.

d. **Error detector** — compares the measured value to the reference point. If the values differ, an error signal is generated.

e. **Controller** — determines the appropriate response. For temperature: thermoregulatory center in the hypothalamus.

f. **Efferent neurons** — carry control signals from the brain to the effectors.

g. **Effectors** — carry out the response. Examples: sweat glands, blood vessels.

h. **System output** — the response changes the regulated variable. Example: sweating and vasodilation lower body temperature.

Exam Question: (2023 Q1): "Which of these statements best describes negative feedback loops that influence homeostasis?" Answer: **c) Efferent nerves are excited, and the system diminishes a particular physical response.**

1.2 Negative Feedback ★

1.2.1 Key Features of Negative Feedback

a. Negative feedback is a control mechanism in which the **response counteracts the initial change** to restore the system toward its set point

b. The system works to return a variable back toward its set point

c. The response always opposes the initial change (stimulus)

d. Maintains stable internal conditions (homeostasis)

1.2.2 Negative Feedback Loop Steps

a. A stimulus causes a deviation from the set point

b. A receptor senses the change

c. The control center compares the value to the set point

d. An effector produces a response

e. The response reduces the original error

1.2.3 Important Notes

a. Negative feedback does **not eliminate** the stimulus, only reduces it

b. Most physiological regulation uses negative feedback

c. Opposite of positive feedback

2 Cell Biology

Cells are the basic structural and functional units of all living organisms.

2.1 Cell Structure

2.1.1 Cell Membrane ★

The cell membrane (plasma membrane) is a **phospholipid bilayer** that surrounds every cell.

a. Structure: two layers of lipids with hydrophobic tails facing inward and hydrophilic heads facing outward

b. **Selectively permeable** — small nonpolar molecules (O_2, CO_2) pass freely; ions and large molecules need channels, carriers, or pumps

c. Hosts proteins — receptors, channels, enzymes, and pumps are embedded in or attached to the membrane

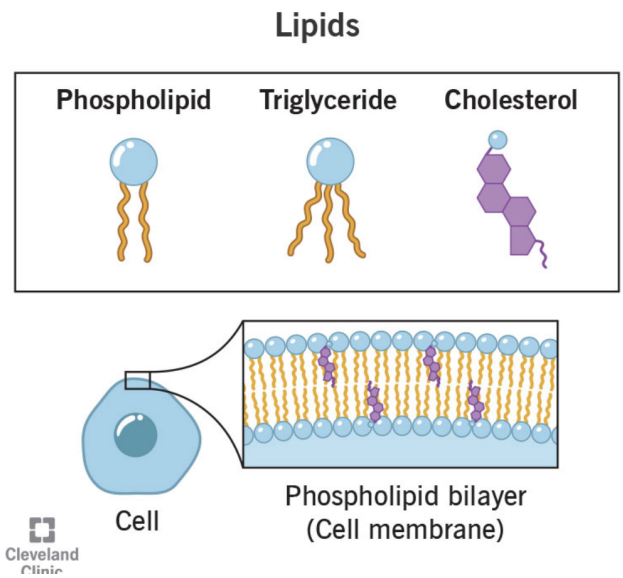

Figure 2: Structure of a phospholipid with hydrophilic head and hydrophobic tails.

Exam Question: (2023 Q4): "Which of the following can't cross a lipid bilayer?" Answer: **a) Ions because they are charged.**

Exam Question: (2023 Q5): "Do steroid molecules need a channel to be transported or diffuse across the cell membrane?" Answer: **b) No, because they are lipid soluble and can pass directly through the phospholipid bilayer.**

2.1.2 Key Spatial Terms

a. **Extracellular** — outside the cell. Na^+ and Cl^- are concentrated here.

b. **Intracellular** — inside the cell. K^+ is concentrated here.

c. **Cytoplasm** — gel-like fluid filling the inside of the cell (excluding the nucleus). Contains organelles, dissolved ions, proteins, and nutrients.

2.1.3 Organelles

a. **Nucleus** — contains DNA; the control centre of the cell. Transcription happens here.

b. **Ribosome** — the site of protein synthesis (translation). Made of rRNA and proteins. Can be free in the cytoplasm or attached to the rough ER.

c. **Mitochondria** — the powerhouse of the cell. Produces ATP through oxidative phosphorylation. Neurons have enormous numbers of mitochondria.

d. **Endoplasmic reticulum (ER)**:

- Rough ER — studded with ribosomes; synthesizes and folds proteins
- Smooth ER — no ribosomes; synthesizes lipids and detoxifies substances

e. **Golgi apparatus** — packages, modifies, and sorts proteins for transport (cell's post office)

f. **Vesicles** — small membrane-bound sacs that transport substances within the cell or release contents outside (exocytosis)

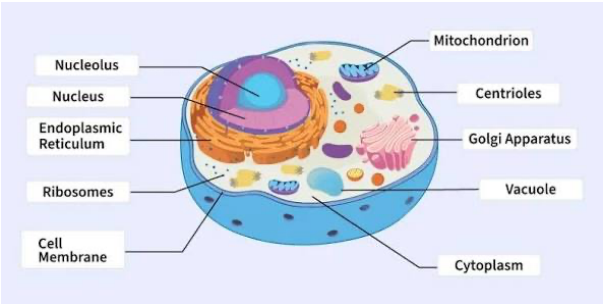

Figure 3: Overview of a typical animal cell showing key organelles.

Exam Question: (2025 MC4): "Which of the following options best describes the function of the ribosome in a cell?" Answer: **c) The ribosomes are like hobby makers using a computer aided design file to create a 3D printed object.**

Exam Question: (2023 Q8): "Which of these statements does NOT apply to ribosomes?" Answer: **c) They are covered by a membrane.**

2.1.4 Hormones

A **hormone** is a chemical messenger produced by a gland or specialized cell and released into the bloodstream. It travels through the blood to act on target cells that have the right receptor.

Important: Hormones are **not RNA**. They are proteins, peptides, or steroids (lipid-based). Hormones are a *result* of gene expression (DNA → mRNA → protein hormone), not part of the process itself.

2.1.5 Tissues

A **tissue** is a group of similar cells that work together to perform a specific function. Cells → tissues → organs → organ systems.

The four primary tissue types:

a. **Epithelial tissue** — covers body surfaces and lines cavities

b. **Connective tissue** — supports, connects, and separates other tissues. Includes bone, cartilage, blood, and the layers wrapping nerves

c. **Muscle tissue** — contracts to produce movement (skeletal, smooth, cardiac)

d. **Nervous tissue** — transmits electrical signals. Made of neurons and glial cells

2.2 Cell Transport ★

2.2.1 Passive Transport (No Energy Required)

Substances move **down** their concentration gradient (high → low).

a. **Simple diffusion** — small nonpolar molecules pass directly through the phospholipid bilayer (e.g., O_2, CO_2)

b. **Facilitated diffusion** — larger or charged molecules move through channels or carriers in the membrane. Still down the gradient, still no energy.

- Channel-mediated — ion channels (e.g., K^+ leak channels)
- Carrier-mediated — a carrier protein binds, changes shape, and releases

c. **Osmosis** — diffusion of water across a selectively permeable membrane

2.2.2 Active Transport (Energy Required) ★

Substances move **against** their concentration gradient (low → high). Requires ATP.

a. **Primary active transport** — a pump directly uses ATP.
Key example: **Na^+/K^+ ATPase pump** — pumps 3 Na^+ out and 2 K^+ in per ATP.

b. **Secondary active transport** — uses the energy stored in an existing ion gradient (created by primary active transport). No direct ATP use, but depends on ATP indirectly.

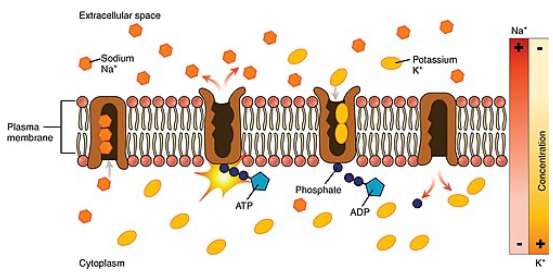

Figure 4: The Na^+/K^+ ATPase pump moves 3 Na^+ out and 2 K^+ in per ATP.

Exam Question: (2024 Q1): "What is the main energy source for the Na^+-K^+ pump in active transport?" Answer: **B) ATP.**

2.2.3 Vesicular Transport

a. **Exocytosis** — vesicles fuse with the cell membrane and release contents outside. Example: neurotransmitter release.

b. **Endocytosis** — the cell membrane wraps around an extracellular substance and pulls it inside.

c. **Cytosis** — general term covering both exocytosis and endocytosis.

Exam Question: (2025 Q13): "Figure 2 displays several molecular transport mechanisms at the neuronal synapse. Label a, b, c, d." Answers: a) Facilitated diffusion, channel mediated; b) Exocytosis; c) Facilitated diffusion, channel mediated; d) Secondary active transport, carrier mediated (reuptake).

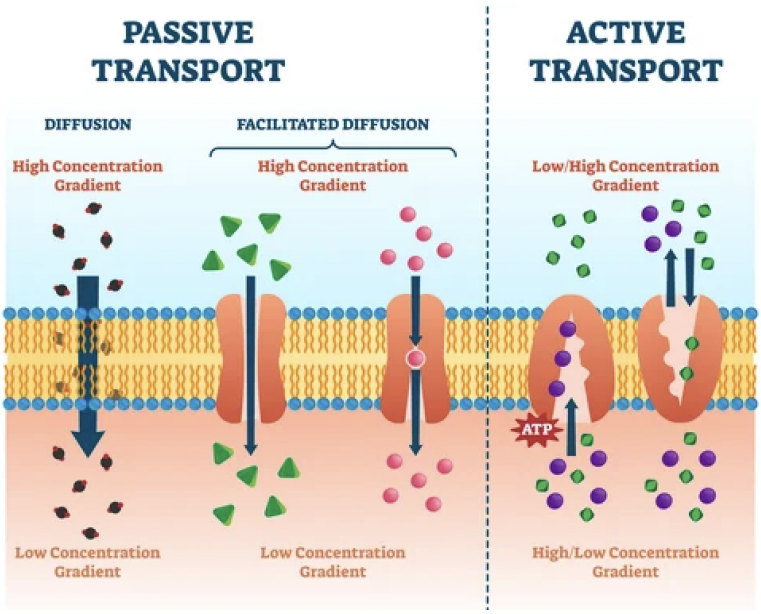

Figure 5: Comparison of passive and active transport mechanisms across the cell membrane.

2.2.4 Flux

Flux is the rate of movement of ions or molecules across a membrane. Net flux depends on:

a. The concentration gradient (chemical driving force)

b. The electrical gradient (for charged particles)

c. The permeability of the membrane to that substance

2.3 Cell Energy ★

2.3.1 ATP

ATP (adenosine triphosphate) is the cell's primary energy currency.

a. Made of adenine + ribose sugar + three phosphate groups

b. Energy is released when the terminal phosphate bond is broken (hydrolysis):

$$\mathrm{ATP} + \mathrm{H_2O} \xrightarrow{\mathrm{ATPase}} \mathrm{ADP} + \mathrm{P}_i + \text{energy}$$

c. ATP is continuously recycled

d. Most ATP is produced by **oxidative phosphorylation** in the mitochondria

2.3.2 Glycolysis and ATP Production Locations ★

a. **Glycolysis** occurs in the **cytosol** (cytoplasm), **NOT** in the mitochondria

b. **Krebs cycle** (citric acid cycle) occurs in the **mitochondrial matrix**

c. **Oxidative phosphorylation** occurs at the **inner mitochondrial membrane** — produces the **most ATP** (this is tested almost every year)

Exam Question: (2025 MC2): "In the game Cytosis, there was a way to get more ATP in a turn if you had an extra glucose token. In real mitochondria, what accounts for the most ATP production?" Answer: **b) Electron transport and oxidative phosphorylation.**

Exam Question: (2024 Q18): "Which process generates the most ATP during cellular respiration?" Answer: **B) Oxidative phosphorylation.**

Exam Question: (2025 MC8): "Where are the enzymes for glycolysis located?" Answer: **c) In the cytosol.**

Exam Question: (2024 Q7/2023 Q7): "Where are the enzymes for glycolysis located?" Answer: **b) In the cytosol.**

Exam Question: (2023 Q6): "Which of the following options best describes the function of the mitochondria in a cell?" Answer: **a) The mitochondria are like the engine of a car, providing energy to power functions of a cell.** (Note: also accepted b — like fuel in a car.)

Tested EVERY year: Know where each stage of cellular respiration happens and that oxidative phosphorylation produces the most ATP. This is a guaranteed question.

3 DNA ★

DNA (Deoxyribonucleic Acid) stores genetic information.

3.1 DNA Building Blocks

DNA is made of repeating units called **nucleotides**.

3.1.1 Nucleotide Components

Each nucleotide has:

a. a phosphate group

b. a deoxyribose sugar

c. a nitrogenous base

Exam Question: (2024 Q33/34): "What is the basic distinguishing structural unit of DNA?" Answer: **B) Nucleotide.** "What makes up a nucleotide?" Answer: **C) A sugar, a phosphate group, and a nitrogen base.**

3.1.2 The Four DNA Bases

A (Adenine), T (Thymine), C (Cytosine), G (Guanine). The sugar and phosphate form the backbone.

3.2 Base Pairing and Direction

3.2.1 Complementary Base Pairing

a. A pairs with T (2 hydrogen bonds)

b. C pairs with G (3 hydrogen bonds)

Exam Question: (2024 Q25): "Which type of bond is found between base pairs in the DNA structure?" Answer: **C) Hydrogen bond.**

Exam Question: (2024 Q5): "Which RNA nucleobase pairs with adenine in DNA?" Answer: **D) Uracil.**

3.2.2 Antiparallel Strands

The two strands run in opposite directions — this is called **antiparallel**.

a. One strand is 5' → 3'

b. The other is 3' → 5'

3.3 Why DNA Has the Same Width (IMPORTANT NOTE)

3.3.1 Base Size Categories

a. **Purines** (A, G) are large, two-ring bases

b. **Pyrimidines** (C, T) are small, one-ring bases

3.3.2 Why Pairing Keeps Width Constant

Base pairing is always Purine + Pyrimidine. This keeps the distance between the two backbones constant.

3.3.3 DNA Right-Hand Twist ★

DNA forms a **right-handed** double helix. To visualize this:

a. Place the **thumb of your right hand** along the axis of the DNA strand

b. Your **curled fingers** follow the direction of the twist (the base pair steps spiral upward to the right)

c. This is the standard B-form DNA that exists in your body

Exam Question: (2025 MC14): "Which is the best way to accurately describe how the DNA molecule twists?" Answer: **a) The DNA twists properly when you place the thumb of your right hand along the axis of the strand and your curled fingers can easily walk along the base pair steps.**

Exam Question: (2023 Q2): "Which of the following analogies best describes how DNA functions?" Answer: **c) DNA is like a series of buttons that can be undone as required to access only the required information, without undoing all buttons.**

3.4 Gene Expression ★

Gene expression is the process by which information in DNA is used to make a protein. It consists of two steps: transcription and translation.

Protein is made of amino acids. Messenger RNA (mRNA) is USED (not composed) as the template to make proteins.

RNA includes introns, mRNA, rRNA, and tRNA. Introns are parts of RNA that are removed before a protein is made. Proteins, enzymes, and hormones are **not** RNA.

3.5 Mutations

A **mutation** is a permanent change in the DNA sequence.

Point mutation — a change in a single nucleotide:

a. **Silent mutation** — new codon still codes for the same amino acid (no effect)

b. **Missense mutation** — new codon codes for a different amino acid

c. **Nonsense mutation** — new codon becomes a stop codon, producing a truncated protein

3.5.1 Transcription ★

Transcription is the process of making RNA from DNA.

a. **Template strand** — read by RNA polymerase in the 3' → 5' direction

b. **Coding strand** — same sequence as RNA (T replaced by U). Not directly read.

c. RNA is synthesized 5' → 3'

Exam Question: (2025 Q1–Q3): Given two complementary DNA strands, identify the coding strand (has start codon ATG), identify it by the start/stop codons and 5' to 3' direction, then write the mRNA transcription. Answer for Q3: 5'–AUG—AUC—CUA—UAU—UGA—CCG—UAG–3'

Exam Question: (2024 Q44): "During transcription, DNA is copied into:" Answer: **A) mRNA.**

Exam Question: (2023 Q1, 2024 Q1, 2025 Q1–3): DNA → mRNA transcription and identifying coding/template strands appear EVERY year.

3.5.2 Precursor mRNA (pre-mRNA)

Precursor mRNA (pre-mRNA) is the rough draft of RNA produced directly from DNA.

a. **Exons** — useful sections that code for protein (kept)

b. **Introns** — non-coding "junk" sections (removed)

c. Processing: splicing (remove introns), 5' cap, 3' poly-A tail

3.5.3 Types of RNA and Their Roles

RNA type	Role	Analogy
mRNA	Carries instructions to ribosome	Recipe
tRNA	Delivers amino acids during translation	Delivery truck
rRNA	Builds the ribosome structure	Factory machinery
pre-mRNA	Unprocessed mRNA (contains introns)	Rough draft

Exam Question: (2023 Q3): "The best comparison of how tRNA interacts with the ribosome to create a protein is how:" Answer: **a) A wandering catering server is to a plated service meal.**

3.5.4 Translation ★

Translation converts RNA into protein.

Stages:

1. **Initiation** — ribosome assembles around start codon (AUG); first tRNA carrying methionine binds
2. **Elongation** — ribosome moves along mRNA 5' → 3'; tRNA delivers amino acids; peptide bonds form
3. **Termination** — ribosome reaches a stop codon (UAA, UAG, or UGA); polypeptide is released

Exam Question: (2024 Q12): "The termination of protein synthesis occurs when the ribosome encounters:" Answer: **B) A stop codon.**

Exam Question: (2024 Q16): "The start codon AUG codes for which amino acid?" Answer: **A) Methionine.**

Exam Question: (2025 Q4): "Using the mRNA, what is the translated protein sequence?" Answer: **Met–Ile–Leu–Tyr** (STOP is not translated).

Exam Question: (2025 Q5): "Assume more RNA processing and one codon could be an exon or intron. What is the translated protein sequence that results in a longer peptide chain?" Answer: **Met–Ile–Leu–Tyr–Pro** (the UGA stop codon is removed as an intron).

3.5.5 Non-Coding and Non-Template DNA

Non-coding DNA refers to regions that do not code for proteins (introns, regulatory sequences). The non-template strand (coding strand) has the same sequence as mRNA with T instead of U.

3.5.6 Important Clarification

DNA replication is **not part of gene expression**. It copies DNA using DNA as the template: DNA → DNA.

Process	Template
DNA replication	DNA
Transcription	DNA
Translation	mRNA

4 Neuron Structure and Function ★

A **neuron** is a specialized cell that transmits electrical signals for communication in the nervous system.

4.1 Neuron Structure

4.1.1 Main Parts

a. **Dendrites** — receive and integrate incoming signals

b. **Cell body (soma)** — contains the nucleus and integrates signals

c. **Axon** — transmits electrical signals away from the cell body

d. **Axon terminals** — release neurotransmitters

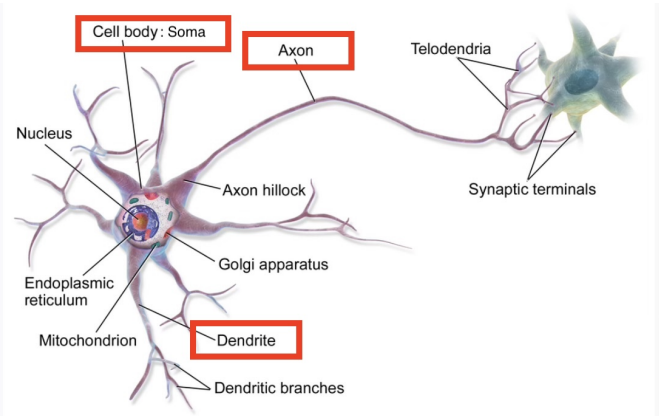

Figure 6: Structure of a neuron showing dendrites, cell body, axon, and axon terminals.

4.2 Key Cellular Components

(Mitochondria, ATPase, Pump, Cell Membrane, Ribosome — as in original notes.)

4.3 Ion Channels ★

4.3.1 Types of Ion Channels

a. **Voltage-gated channels** — open when membrane voltage changes

b. **Ligand-gated channels** — require a chemical ligand to open

c. **Leak channels** — always open; responsible for maintaining RMP

Exam Question: (2024 Q3): "Which type of gated ion channel is crucial for the propagation of action potentials?" Answer: **C) Voltage-gated.**

Exam Question: (2024 Q42): "Which type of ion channel opens or closes in response to a change in membrane potential?" Answer: **B) Voltage-gated channels.**

4.3.2 Resting Membrane Potential (RMP) ★

The **resting membrane potential** (RMP) is approximately **−70mV** — the inside of the neuron is negative relative to the outside.

Ion concentrations (standard values):

Ion	Inside (mM)	Outside (mM)	Direction at rest
K^+	115	2	Wants OUT (chemical)
Na^+	12	150	Wants IN (chemical)
Cl^-	8	150	Wants IN (chemical)

Why K^+ sets the RMP: At rest, K^+ leak channels are always open. K^+ is driven OUT by the chemical force. As K^+ leaves, large negatively-charged proteins trapped inside make the inside more negative. K^+ keeps leaving until the electrical force pulling it back in exactly balances the chemical force pushing it out.

Exam Question: (2024 Q2): "Resting membrane potential is approximately:" Answer: **B) −70 mV.**

The Nernst Equation — calculates the equilibrium potential for a single ion:

$$E_X = \frac{61}{z} \log \frac{[X]_{out}}{[X]_{in}}$$

Exam Question: (2024 Q23): "What does the Nernst equation calculate?" Answer: **B) The equilibrium potential for an ion.**

Exam Question: (2024 Q31): "If $[K^+]_{out} = 5$ mMol/L and $[K^+]_{in} = 150$ mMol/L, what is E_{K^+}?" Answer: **B) −90 mV.** Calculation: $E_{K^+} = \frac{61}{+1} \log \frac{5}{150} = 61 \times (-1.477) = -90.1$ mV.

The Goldman Equation (GHK):

$$V_m = 61 \log \frac{P_{K^+}[K^+_{out}] + P_{Na^+}[Na^+_{out}] + P_{Cl^-}[Cl^-_{in}]}{P_{K^+}[K^+_{in}] + P_{Na^+}[Na^+_{in}] + P_{Cl^-}[Cl^-_{out}]}$$

where $P_{K^+} = 1$, $P_{Na^+} = 0.04$, $P_{Cl^-} = 0.3$.

Exam Question: (2025 Q6): GABA/Cl^-/K^+ calculation using Goldman & Nernst equations with old neuron data. This is the biggest calculation question — appears EVERY year. Know how to calculate V_m and E_{ion} and explain which direction the ion moves.

Na^+/K^+ ATPase Pump:

a. Uses ATP to pump 3 Na^+ out and 2 K^+ in per cycle

b. Maintains the concentration gradients

c. If the pump stops: gradients disappear → RMP collapses toward 0 mV → neuron cannot fire

Hyperkalemia: Elevated extracellular K^+ → reduced K^+ gradient → RMP becomes less negative (depolarized) → initially hyperexcitable, then unexcitable → muscle weakness, cardiac arrhythmias.

Exam Question: (2025 MC9): "For a given ion crossing a cell membrane, the electrical driving force:" Answer: **b) Depends only on the electrical force, regardless of the concentration gradient of the ion.**

Exam Question: (2023 Q9): "For a given molecule crossing a cell membrane, the chemical driving force:" Answer: **b) Depends only on the concentration gradient, regardless of whether or not the molecule is an ion.**

4.4 Action Potential ★

An **action potential** is a brief electrical signal — a rapid, temporary reversal of membrane potential — that travels along a neuron to transmit information.

4.4.1 When Does It Happen?

At rest, the neuron sits at RMP = −70 mV. An action potential only occurs if the membrane is depolarized enough to reach the **threshold potential** (≈ −55 to −60 mV).

All-or-nothing law:

a. If threshold is reached → a full action potential fires

b. If threshold is not reached → nothing happens (no partial AP)

Exam Question: (2025 Q11): "Choose the best analogy for the action potential pattern." Answers A, C, or D accepted with justification. **C (air pistol) is best** — "All or nothing." B (dimmer switch) was NOT accepted because there is nothing "in between."

Exam Question: (2025 MC15): "Suppose the net summation brings the membrane potential to −60 mV. Will an action potential occur?" Answer: **a) No, an action potential will not occur** (threshold is −55 mV, −60 mV is below threshold).

4.4.2 Phases of the Action Potential ★

Phase	Na^+ channels	K^+ channels	Result
Resting (−70 mV)	Closed	Closed	Stable RMP
Depolarization	Open	Closed	V_m rises rapidly (Na^+ rushes in)
Repolarization	Inactivated	Open	V_m falls (K^+ rushes out)
Hyperpolarization	Inactivated	Slowly closing	V_m dips below RMP
Recovery	Closed (reset)	Closed	Returns to RMP

Exam Question: (2023 Q9 / 2024 Q9): "What term describes what is happening at point 'a' [the peak]?" Answer: **Depolarization**; main cause: **Na^+ influx through voltage-gated Na^+ channels.**

Exam Question: (2023 Q10 / 2024 Q10): "What term describes point 'b' [the downslope]?" Answer: **Repolarization**; main cause: **K^+ efflux through voltage-gated K^+ channels.**

Exam Question: (2024 Q32): "During depolarization, which ion channels open first?" Answer: **B) Sodium channels.**

4.4.3 Refractory Period

1. **Absolute refractory period** — neuron cannot fire at all. Na^+ channels are inactivated.
2. **Relative refractory period** — can fire, but only with a stronger stimulus. K^+ channels still partially open.

Exam Question: (2023 Q11 / 2024 Q11): "There are two periods labelled 'i' and 'ii'. What are they called?" Answer: i = **Absolute refractory period**; ii = **Relative refractory period.**

4.4.4 Propagation of the Action Potential

1. **Contiguous conduction** — continuous along unmyelinated axons. **Slow**.
2. **Saltatory conduction** — "jumps" between nodes of Ranvier in myelinated axons. **Much faster** and more energy-efficient.

Exam Question: (2024 Q36): "Saltatory conduction occurs in:" Answer: **B) Myelinated axons.**

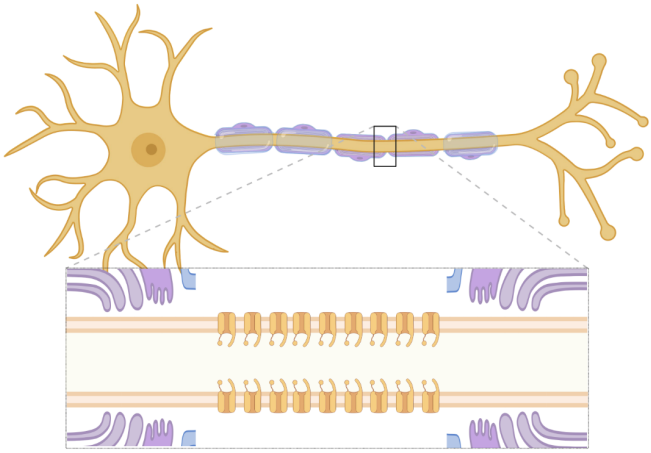

Figure 9: Contiguous vs saltatory conduction along unmyelinated and myelinated axons.

4.4.5 Properties of Action Potentials

a. **All-or-nothing**

b. **Non-decrementing** (does not weaken with distance)

c. **Unidirectional** (absolute refractory period behind)

d. Speed depends on axon diameter (larger = faster) and myelination (myelinated = faster)

Exam Question: (2023 Q11): "Which factor would increase the speed of nerve impulses?" Answer: **c) Increasing axon diameter because there is a greater surface area.**

4.4.6 Oligodendrocytes vs Schwann Cells ★

Both produce **myelin**, but in different locations:

a. **Schwann cells** — produce myelin in the **PNS** (peripheral nervous system). Each Schwann cell wraps around **one** segment of **one** axon.

b. **Oligodendrocytes** — produce myelin in the **CNS** (brain and spinal cord). Each oligodendrocyte can wrap around segments of **multiple** axons.

Exam Question: (2024 Q6): "Which type of cell in the CNS creates myelin sheaths around axons?" Answer: **C) Oligodendrocytes.**

Exam Question: (2024 Q14): "What is the role of myelin sheath in neurons?" Answer: **C) Speeding up signal transmission.**

Exam Question: (2025 MC5): "An Alpha fiber axon is to Schwann Cell as a coffee cup is to:" Answer: **b) The corrugated cardboard that forms a sleeve.**

4.5 Synapse ★

4.5.1 Main Components of a Synapse

a. Axon terminal — contains vesicles filled with neurotransmitter

b. Presynaptic neuron — sends the signal

c. Synaptic cleft — small gap between neurons

d. Postsynaptic cell — receives the signal

e. Postsynaptic receptors — bind neurotransmitters, open ion channels

f. Postsynaptic potential — EPSP or IPSP

Exam Question: (2024 Q24): "The synaptic cleft is the gap between:" Answer: **B) A presynaptic neuron and a postsynaptic neuron in a chemical synapse.**

4.5.2 Types of Synapses

1. **Chemical Synapse** — uses neurotransmitters across synaptic cleft
2. **Electrical Synapse** — uses gap junctions for direct ion flow. Very fast, no delay.

Step-by-step chemical synapse process:

1. AP arrives at axon terminal
2. Voltage-gated Ca^{2+} channels open → Ca^{2+} rushes in
3. Ca^{2+} causes vesicle fusion (exocytosis) → neurotransmitter released
4. neurotransmitter diffuses across synaptic cleft
5. neurotransmitter binds to postsynaptic receptors → ion channels open
6. Postsynaptic potential created (EPSP or IPSP)

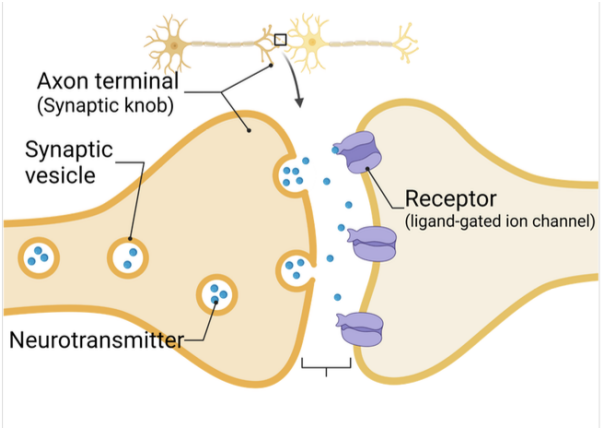

Figure 10: Chemical synapse showing axon terminal, synaptic vesicles, cleft, and postsynaptic receptors.

Exam Question: (2024 Q21): "What happens when an action potential reaches the presynaptic terminal?" Answer: **A) It triggers the release of neurotransmitters.**

Exam Question: (2024 Q40): "Which ion's influx is crucial for the release of neurotransmitters?" Answer: **C) Calcium (Ca^{2+}).**

Exam Question: (2025 MC6): "What is the most immediate action missing in Cytosis when you perform exocytosis?" Answer: **a) Ca^{2+} because it is needed for proteins to fuse vesicles with the plasma membrane.**

4.5.3 EPSP vs. IPSP vs. GPSP ★

Property	EPSP	IPSP	GPSP
Direction	Depolarizing	Hyperpolarizing	Either
Caused by	Na^+ entering	Cl^- entering / K^+ leaving	—
Effect	More likely to fire	Less likely to fire	Sum determines
All-or-nothing?	No (graded)	No (graded)	No (graded)

Exam Question: (2024 Q22): "EPSP increases the likelihood of:" Answer: **A) An action potential.**

Exam Question: (2024 Q7): "Hyperpolarization of a neuron makes it:" Answer: **A) Less likely to fire an action potential.**

GABA (γ-aminobutyric acid) is the main **inhibitory** neurotransmitter in the brain. GABA binds → Cl^- channels open → Cl^- moves inward → hyperpolarization → inhibition.

Glutamate is the main **excitatory** neurotransmitter in the brain. Glutamate binds → Na^+ channels open → depolarization (EPSP).

Glycine is an **inhibitory** neurotransmitter in the spinal cord and brainstem.

4.5.4 Types of Receptors ★

1. **Ionotropic receptor** (ligand-gated ion channel) — receptor IS an ion channel; neurotransmitter binds → channel opens immediately. **Fast**.

2. **Metabotropic receptor** (G-protein coupled receptor, GPCR) — receptor is NOT an ion channel; activates G-protein → second messenger. **Slower, longer lasting.**

4.5.5 Acetylcholine (ACh) ★

1. **Nicotinic receptor** — **ionotropic** (ligand-gated). Found at neuromuscular junction. Fast: ACh binds → Na^+ enters → depolarization → muscle contracts.

2. **Muscarinic receptor** — **metabotropic** (G-protein coupled). Found in parasympathetic NS. Slower, indirect via second messenger.

Exam Question: (2025 MC7): "Nicotinic and muscarinic receptors are two kinds of adrenergic receptors." Answer: **a) False, they are cholinergic receptors.**

Exam Question: (2024 Q9): "What neurotransmitter is always associated with muscle activation at the motor endplate?" Answer: **D) Acetylcholine.**

Exam Question: (2025 Q14): "List two more common terms for the postsynaptic receptor (structure 'c')." Acceptable: non-specific cation channel, ionotropic receptor, cholinergic receptor, ligand-gated ion channel.

4.6 Synaptic Integration ★

A single neuron receives signals from many neurons simultaneously. Each produces an EPSP or IPSP. The neuron sums all of these to decide whether to fire.

4.6.1 Temporal Summation

Same presynaptic neuron fires repeatedly in rapid succession. EPSPs accumulate before previous ones decay.

4.6.2 Spatial Summation

Multiple different presynaptic neurons fire at the same time, each contacting the same postsynaptic neuron.

Exam Question: (2025 Q10 / 2024 Q10 / 2023 Q13): "If the selection of responses are examples of neuronal integration, what could cause the different responses?" Answer: **Temporal summation** and **Spatial summation.** — Tested EVERY year.

Exam Question: (2024 Q47): "The Grand Post-Synaptic Potential (GPSP) can be affected by:" Answer: **A) Temporal summation.**

Exam Question: (2023 Q13): "Which statement applies to receptor potentials?" Answer: **a) They occur in neuronal cell bodies upon summation of presynaptic input.**

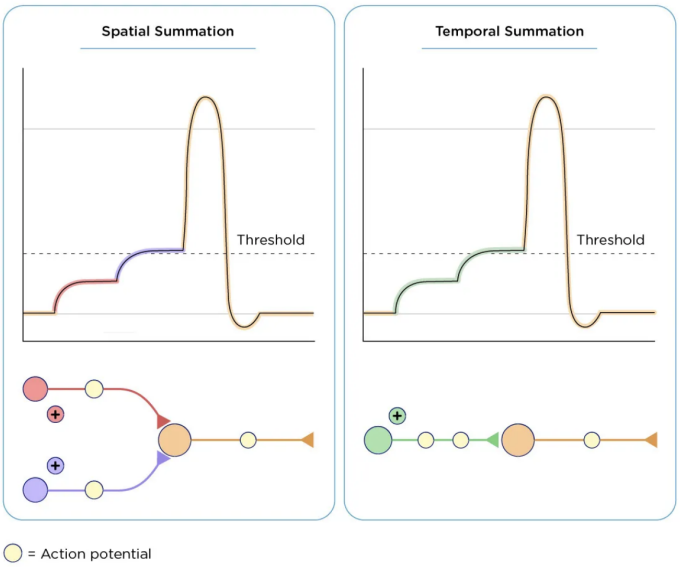

Figure 11: Spatial and temporal summation of EPSPs at the postsynaptic neuron.

4.6.3 Lateral Inhibition ★

Lateral inhibition is when an excited neuron **inhibits its neighbours** to sharpen sensory perception and increase contrast.

a. The most strongly stimulated neuron fires and simultaneously sends inhibitory signals to adjacent neurons

b. This suppresses weaker signals around it, making the strongest signal stand out

c. Result: enhanced precision of sensory perception (e.g., sharper touch, clearer edges in vision)

Exam Question: (2024 Q27): "What mechanism enhances the precision of sensory perception by reducing the activity of neighboring neurons?" Answer: **C) Lateral inhibition.**

4.6.4 Presynaptic Inhibition

Presynaptic inhibition occurs when an **inhibitory neuron synapses directly onto a presynaptic terminal** (not onto the postsynaptic neuron).

a. The inhibitory neuron reduces the amount of neurotransmitter the presynaptic neuron can release

b. The postsynaptic neuron receives a weaker signal

c. This is different from postsynaptic inhibition (IPSP), which acts on the postsynaptic neuron directly

Exam Question: (2024 Q37): "Presynaptic inhibition occurs when:" Answer: **D) An inhibitory neuron synapses directly with a presynaptic neuron.**

4.6.5 How the Signal Ends

1. **Reuptake** — presynaptic terminal takes neurotransmitter back up through transporter proteins

2. **Enzymatic degradation** — enzymes break neurotransmitter apart (e.g., acetylcholinesterase breaks ACh)
3. **Diffusion** — neurotransmitter drifts away from the synapse

4.7 Central Nervous System (CNS)

The **CNS** consists of the brain and spinal cord. Integration hub.

4.7.1 The Brain

Key regions:

a. **Cerebral cortex** — outer grey matter; neuron cell bodies, integration

b. **Frontal lobe** — voluntary motor control, planning, decision-making

c. **Somatosensory cortex** — receives touch, pressure, pain, temperature

d. **Brainstem** — midbrain + pons + medulla. Controls vital automatic functions

e. **Cerebellum** — coordinates movement and balance

4.7.2 Grey vs White Matter

a. **Grey matter** — neuronal cell bodies. Site of synaptic integration.

b. **White matter** — myelinated axons. Signal transmission.

4.7.3 Spinal Cord

a. **Dorsal horn** — sensory processing

b. **Ventral horn** — motor neuron cell bodies

c. **Dorsal root** — afferent (sensory) fibers enter

d. **Ventral root** — efferent (motor) fibers exit

Bell–Magendie Law: Dorsal = sensory, Ventral = motor.

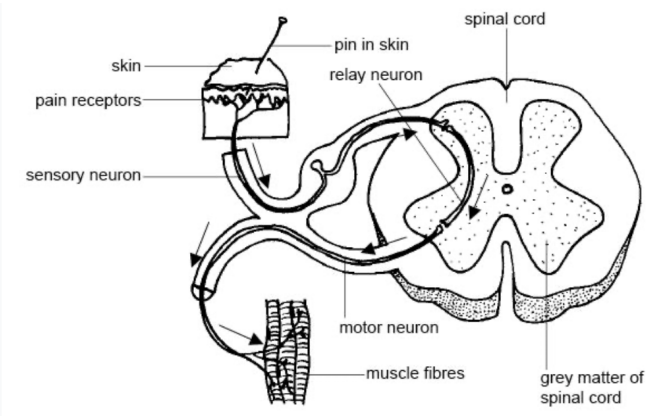

Figure 15: Spinal cord cross-section showing dorsal/ventral horns and roots.

Exam Question: (2024 Q46): "Which part of the spinal cord contains motor neurons?"
Answer: **C) Ventral horn.**

4.7.4 Ascending vs Descending Tracts

a. **Ascending tracts** — carry sensory information to the brain

b. **Descending tracts** — carry motor commands from the brain

Exam Question: (2025 Q17): "The postsynaptic neuron travels from this level of the spinal cord to the brain. What is this pathway called?" Answer: **Ascending tract** (also: Sensory pathway, Afferent pathway, Somatosensory pathway).

5 Nervous System Overview & Reflexes ★

5.1 Neurons: The Longest-Lived Cells

Neurons are among the longest-lived cells — many survive from birth to death without being replaced. The brain consumes ∼20% of the body's energy at rest.

5.2 Divisions of the Nervous System

1. **Central Nervous System (CNS)** — brain + spinal cord
2. **Peripheral Nervous System (PNS)** — all nerves outside the CNS
   a. Afferent (sensory) division — carries signals TO the CNS
   b. Efferent (motor) division — carries signals FROM the CNS
      - Somatic NS → skeletal muscle (voluntary)
      - Autonomic NS → smooth muscle, cardiac muscle, glands (involuntary)

Exam Question: (2024 Q15): "The brain and spinal cord are part of the:" Answer: **B) Central Nervous System.**

Exam Question: (2024 Q12): "What is the function of afferent neurons?" Answer: **A) To transmit messages from receptors to the spinal cord and brain.**

Exam Question: (2024 Q50): Mnemonic: **SAME** (Sensory Afferent, Motor Efferent).

5.3 Receptor Physiology ★

Sensory receptors convert stimuli from the environment into electrical signals the CNS can process. The CNS extracts **four key attributes** from every sensory signal: **Modality**, **Intensity**, **Locality**, and **Duration**.

5.3.1 Adequate Stimulus and Receptor Types

Every receptor has an **adequate stimulus** — the specific type of energy it is most sensitive to.

a. **Mechanoreceptors** — respond to mechanical deformation (touch, pressure, vibration, stretch)

b. **Thermoreceptors** — respond to temperature changes

c. **Nociceptors** — respond to pain / tissue damage

d. **Photoreceptors** — respond to light

e. **Chemoreceptors** — respond to chemicals (taste, smell, blood O_2)

Exam Question: (2024 Q10): "Ion channels that open and close in response to mechanical loading are associated with:" Answer: **A) Mechanoreceptors.**

5.3.2 Receptor (Generator) Potential

When a receptor is stimulated, it produces a **receptor potential** (also called **generator potential**) — a local, graded depolarization of the receptor membrane.

a. The receptor potential is **graded** — its amplitude is proportional to stimulus strength

b. It is **not** an action potential — it does not follow the all-or-nothing law

c. If the receptor potential is large enough, it triggers action potentials in the associated afferent neuron

d. Larger receptor potential → higher AP frequency → brain perceives stronger stimulus

5.3.3 1. Modality — "What kind of stimulus is it?"

Modality refers to the type of sensation perceived (touch, pain, heat, light, sound, etc.).

Labeled Line Principle: Each receptor type has its own dedicated neural pathway to a specific region of the brain. The brain determines modality based on **which pathway** is activated, not by the signal itself. Example: activating photoreceptors always produces a sensation of light, even if stimulated mechanically (seeing stars when you rub your eyes).

5.3.4 2. Intensity — "How strong is the stimulus?" ★

The CNS encodes intensity using **frequency coding**:

a. Stronger stimulus → larger receptor potential → **higher frequency** of action potentials

b. Weaker stimulus → smaller receptor potential → lower AP frequency

c. APs are all-or-nothing, so the **size** of each AP is always the same — only the **frequency** changes

Exam Question: (2024 Q29): "The larger the receptor potential, the:" Answer: **B) Higher the frequency of action potentials.**

Key concept: Intensity is NOT encoded by bigger action potentials. It is encoded by **more frequent** action potentials. This is frequency coding.

5.3.5 3. Locality — "Where is the stimulus?"

The CNS determines location using **receptive fields**:

a. A **receptive field** is the area of skin (or other tissue) served by a single sensory neuron

b. **Smaller receptive fields** = better spatial resolution (e.g., fingertips)

c. **Larger receptive fields** = poorer spatial resolution (e.g., back)

d. **Two-point discrimination**: the minimum distance at which two touches are felt as separate points — smaller distance = smaller receptive fields = better resolution

e. **Lateral inhibition** sharpens locality by suppressing activity in neighbouring neurons (see Section 4.6.1)

5.3.6 4. Duration — "How long does the stimulus last?" ★

Duration is encoded by how long the receptor continues to fire. This depends on **receptor adaptation**:

a. **Rapidly adapting (phasic) receptors** — stop responding quickly to a sustained stimulus. They detect **changes** (on/off). Example: Pacinian corpuscle (vibration), Meissner's corpuscle (light touch).

b. **Slowly adapting (tonic) receptors** — continue firing as long as the stimulus is present. They signal **ongoing** stimuli. Example: Merkel cells (pressure), nociceptors (pain).

Exam Question: (2024 Q19): "What type of receptor adapts rapidly to a sustained stimulus?" Answer: **D) Phasic receptors.**

Exam Question: (2024 Q28): "Phasic receptors are characterized by:" Answer: **B) Rapid adaptation to a sustained stimulus.**

Exam Question: (2023 Q15): "Which of these receptors is rapidly adapting?" Answer: **a) Phasic receptor.**

Exam tip: The four sensory attributes framework (Modality, Intensity, Locality, Duration) is a major L10 topic. Know frequency coding for intensity, receptive fields for locality, and phasic vs tonic for duration.

5.4 Somatic Nervous System (SNS)

Controls voluntary movement of skeletal muscle.

a. One neuron runs directly from CNS to muscle (no relay station)

b. Neurotransmitter released: **Acetylcholine (ACh)**

c. Muscle has **nicotinic receptors** (ionotropic, fast)

d. Always excitatory

Exam Question: (2025 Q18): "What type of neuron is shown in row a?" Answer: **Somatic Neuron** (also: Motor neuron).

5.5 Reflexes ★

A **reflex** is a fast, involuntary, stereotyped response processed at the spinal cord level.

5.5.1 Reflex Arc — The Basic Circuit

1. Receptor — senses the stimulus
2. Afferent (sensory) neuron — carries signal to spinal cord via dorsal root
3. Integration centre (spinal cord) — interneurons process the signal
4. Efferent (motor) neuron — carries motor command via ventral root
5. Effector — muscle or gland that produces the response

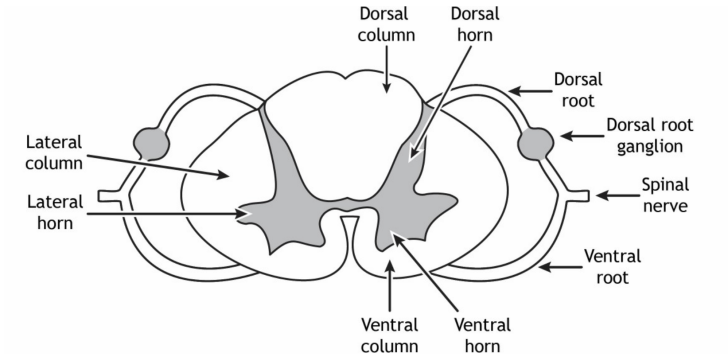

Figure 17: Spinal cord reflex arc showing the complete pathway from receptor to effector.

Exam Question: (2024 Q48): "Reflex arcs do NOT include which component?" Answer: **D) Hormone.**

Exam Question: (2025 Q15): "What is the best label for number 7 [in spinal cord diagram]?" Answer: **Dorsal root ganglion.**

Exam Question: (2025 Q16): "What kind of neuron is labelled number 8?" Answer: **Inhibitory interneuron.** "Is it excited or inhibited?" Answer: **It is excited.**

5.5.2 Withdrawal Reflex

Triggered by painful stimulus (e.g., hot stove). Flexor muscles contract, pulling limb away. Interneurons inhibit antagonist extensor motor neurons (reciprocal inhibition).

5.5.3 Crossed Extensor Reflex

Accompanies withdrawal reflex to maintain balance. While stimulated limb flexes, the opposite limb extends.

5.5.4 Monosynaptic vs Polysynaptic Reflexes ★

a. **Monosynaptic reflex** — only **one synapse** between the sensory and motor neuron (no interneuron). Example: knee-jerk (patellar) reflex. Fastest possible reflex.

b. **Polysynaptic reflex** — **two or more synapses**; involves interneurons. Example: withdrawal reflex (touching hot stove). Slower but more complex.

c. Reflex arcs can be **both** monosynaptic and polysynaptic.

Exam Question: (2024 Q49): "Reflex arcs may be:" Answer: **D) Both A and B** (both monosynaptic and polysynaptic).

Exam Question: (2023 Q12): "In comparing a monosynaptic reflex to an email exchange, which is the most accurate?" Answer: **c) A monosynaptic reflex is like an automatic reply email.**

5.5.5 Why Interneurons and Inhibition Are Essential

Interneurons can both excite and inhibit motor neurons. Reciprocal inhibition ensures opposing muscle groups don't fight each other.

6 Autonomic Nervous System (ANS) ★

The **ANS** controls involuntary functions: smooth muscle, cardiac muscle, and glands.

Pathway structure: Unlike somatic (1 neuron), the ANS uses **two neurons in series**:

1. **Preganglionic neuron** cell body in CNS; axon travels to a ganglion
2. **Postganglionic neuron** cell body in ganglion; axon travels to target organ

Exam Question: (2024 Q11): "The autonomic nervous system typically has a chain of how many neurons from the CNS to the effector organ?" Answer: **B) Two.**

Exam Question: (2024 Q8): "The autonomic nervous system is divided into which two primary systems?" Answer: **B) Sympathetic and Parasympathetic.**

Exam Question: (2025 MC1): "Choose the part of the ANS that represents the place where they wait for the bus [transit analogy]." Answer: **a) The chain ganglion.**

6.1 Sympathetic Division ("Fight or Flight")

Activated during stress, exercise, or danger. Prepares the body for rapid action. The sympathetic division is also called the **thoracolumbar** division because its preganglionic neurons originate from the **thoracic (T1–T12)** and **lumbar (L1–L2)** segments of the spinal cord.

a. Increases heart rate and blood pressure.

b. **Dilates** pupils — dilation means widening/enlarging the opening, letting in more light so you can see threats better.

c. Dilates airways (bronchodilation) to deliver more oxygen.

d. Reduces digestive activity (non-essential during an emergency).

e. Primary postganglionic neurotransmitter: **Norepinephrine (NE)** (also called noradrenaline) — acts on **adrenergic receptors** on target organs. Because it releases norepinephrine, the sympathetic postganglionic neuron is called **adrenergic** ("releases adrenaline-like substances").

Primary postganglionic neurotransmitter (most targets): Norepinephrine (NE) → acts on **adrenergic** receptors (alpha or beta)

The Adrenal Medulla: The **adrenal medulla** is a modified sympathetic ganglion located on top of each kidney.

a. A preganglionic neuron synapses directly onto **chromaffin cells** (instead of a postganglionic neuron)

b. Chromaffin cells release **epinephrine** (∼80%) and **norepinephrine** (∼20%) directly into the **bloodstream**

c. This amplifies the fight or flight response body wide since the hormones reach many organs via blood

d. The adrenal medulla is a **neuroendocrine** organ: nervous input but endocrine hormone output

Exam Question: (2023 Q20): "Which spinal nerves do sympathetic fibres emanate from?" Answer: **a) Thoracic and lumbar levels.**

6.2 Parasympathetic Division ("Rest and Digest")

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Dominant during rest and recovery. Think of it as the calming counter to the sympathetic system.

a. Decreases heart rate.

b. Stimulates digestion (increases gut motility and secretion).

c. Constricts pupils.

d. Primary postganglionic neurotransmitter: **Acetylcholine (ACh)** acting on **muscarinic receptors** (metabotropic, slower) on target organs. Because it releases ACh, the parasympathetic postganglionic neuron is called **cholinergic** ("releases acetylcholine").

The parasympathetic division is also called the **craniosacral** division because its preganglionic neurons originate from the **brainstem (cranial nerves III, VII, IX, X)** and **sacral (S2–S4)** segments of the spinal cord.
Dual Innervation: Most organs receive innervation from **both** sympathetic and parasympathetic divisions. The two divisions often have **opposing effects** on the same organ (example: sympathetic increases heart rate, parasympathetic decreases it). The balance between the two sets the organ activity level.

Exam Question: (2024 Q45): "Which type of receptors are associated with the parasympathetic nervous system?" Answer: **D) Cholinergic.**

Exam Question: (2024 Q20): "Adrenergic receptors in the ANS are activated by:" Answer: **A) Norepinephrine.** (Also accepted: C) Epinephrine.)

Exam Question: (2023 Q17): "Which statement best describes parasympathetic postganglionic fibres?" Answer: **c) They are cholinergic fibres.**

6.3 Receptor Summary for the ANS ★

Cholinergic vs. Adrenergic, plus receptor subtypes:

a. **Cholinergic** neurons release **acetylcholine (ACh)**. This includes: **all** preganglionic neurons (both divisions), **all** parasympathetic postganglionic neurons, and somatic motor neurons.

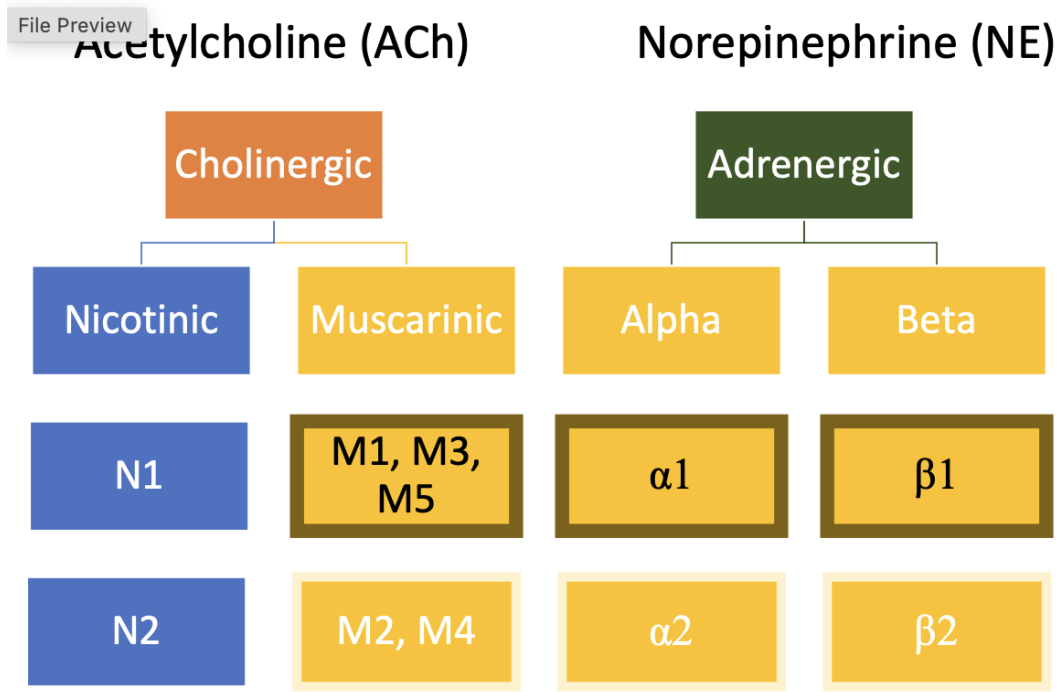

Figure 1: Receptor families in the ANS: cholinergic (ACh) versus adrenergic (NE), with major subtypes.

b. **Adrenergic** neurons release **norepinephrine (NE)**. This includes: **most** sympathetic postganglionic neurons.

c. **Nicotinic receptors** are **ionotropic** (fast) and are found at: all autonomic ganglia (postganglionic neuron cell body or dendrites), and the neuromuscular junction.

   i. **N1:** neuromuscular junction (somatic)

   ii. **N2:** autonomic ganglia (ANS)

d. **Muscarinic receptors** are **metabotropic** (slower) and are found at: parasympathetic postganglionic → target organ synapses.

   i. **M1, M3, M5:** typically excitatory

   ii. **M2, M4:** typically inhibitory (classic: **M2 slows the heart**)

e. **Adrenergic receptors** respond to **NE** (and often **E** as well) and are:

   i. α **receptors:** $\alpha 1$ typically excitatory, $\alpha 2$ typically inhibitory

   ii. β **receptors:** $\beta 1$ typically increases heart activity, $\beta 2$ typically relaxes smooth muscle (classic: bronchodilation)

Ganglion and fibre facts :

a. **Fibre length pattern:** Sympathetic usually has **short preganglionic** and **long postganglionic** fibres, while parasympathetic usually has **long preganglionic** and **short postganglionic** fibres

b. **Myelination:** Preganglionic fibres are lightly myelinated, postganglionic fibres are typically unmyelinated

c. **Effect direction:** ANS effects can excite or inhibit depending on receptor subtype and target tissue

Exam Question: (2025 MC13 / 2024 Q13 / 2023 Q18): "All postganglionic neurons have _____ on their dendrites or cell bodies." Answer: **a) Nicotinic receptors.** Tested every year.

Exam Question: (2023 Q19): "Which best describes nicotinic receptors?" Answer: **b) They respond to acetylcholine released from both sympathetic and parasympathetic preganglionic fibres.**

Exam Question: (2025 Q19): "Referring to row b, what type of receptor is attached to the effector cell?" Answer: **Muscarinic receptor** (cholinergic).

Exception to the rule: sympathetic cholinergic (sweat glands)

a. Most sympathetic postganglionic neurons are **adrenergic** (NE)

b. **Exception:** some sympathetic postganglionic neurons release **ACh** onto **muscarinic** receptors, classically for **sweat glands**

Exam Question: (2025 Q20): "You discover a new nerve that has qualities of both row b and c (depicted in row d targeting sweat gland). How is this nerve the same but different?" Answer: **Sympathetic nerve that releases acetylcholine (ACh)**.

Exam Question: (2025 MC10): "Motor neurons create the fastest reflexes because they:" Answer: **c) Have large diameter axons and target skeletal muscles have nicotinic receptors.**

6.4 Somatic vs. Autonomic Pathway Comparison

Feature	Somatic	Autonomic
Neurons in chain	1	2 (pre and postganglionic)
Target	Skeletal muscle	Smooth and cardiac muscle, glands
Control	Voluntary	Involuntary
Neurotransmitter at target	ACh (nicotinic, N1)	ACh (muscarinic) or NE (adrenergic)
Effect	Always excitatory	Can excite or inhibit

6.5 Cranial Nerves (Overview)

12 pairs of cranial nerves arise from the brainstem and supply head, neck, and some visceral organs.

Key: The **Vagus nerve (CN X)** is the major parasympathetic nerve — heart, lungs, gut.

6.6 Structure of a Peripheral Nerve

Memory trick (inside to outside): **Every Person Eats** — Endoneurium → Perineurium → Epineurium.

Layer	What it wraps	Position
Endoneurium	Individual axon (+ myelin)	Innermost
Perineurium	One fascicle (bundle of axons)	Middle
Epineurium	Entire nerve (all fascicles)	Outermost

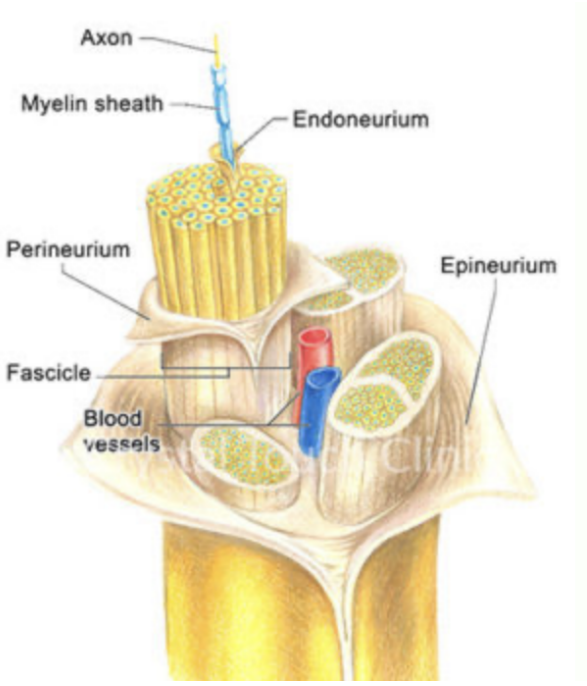

Figure 19: Structure of a peripheral nerve showing endoneurium, perineurium, and epineurium.

7 Thermodynamics in Biology

7.1 Gibbs Free Energy ★

ΔG determines whether a reaction occurs spontaneously.

$$\Delta G = \Delta H - T\Delta S$$

a. $\Delta G < 0$ (exergonic): spontaneous. Example: ATP hydrolysis.

b. $\Delta G > 0$ (endergonic): requires energy. Example: protein synthesis, Na^+/K^+ pump.

c. $\Delta G = 0$: equilibrium.

For ion transport across membranes:

$$\Delta G_{Chem} = RT \ln \frac{[X]_{out}}{[X]_{in}} \qquad \Delta G_{Elec} = ZFV_m$$

where $R = 8.314$ J/mol·K, $T = 309.65$ K, Z = ion valence, $F = 96485$ C/mol, V_m in Volts.

Exam Question: (2025 Q7): "Calculate the Gibbs free energy and explain how it supports your answer." For $GABA_A$ (Cl^-): $\Delta G_{Chem} = 8.314 \times 309.65 \times \ln(150/8) = +7579$ J/mol; $\Delta G_{Elec} = (-1)(96485)(-0.0725) = +6995$ J/mol; $\Delta G_{total} = \Delta G_{Chem} + \Delta G_{Elec}$. **Positive ΔG outward means Cl^- moves inward (not spontaneous outward, so spontaneous inward).**

Exam Question: (2025 Q8): "The old neuron cell is treated with a drug that blocks Cl^- leak channels and V_m becomes slightly depolarized. Does the cell actively transport Cl^-?" Answer: **Yes, active transport of Cl^- outward.** Explanation: net Cl^- was diffusing in passively, balanced by active transport out. Lower [Cl^-] in old neurons (8) vs young neurons (20) is evidence for active Cl^- export.